

VANGUARD

One picture. Every source. Zero delay.

Live Demo Runbook

Multi-Source Defence Situational Awareness System
Problem ID D-05 · HackHertz 2026 Defense Track
Team Destroyer of Worlds · Document version 1.1

Runtime: 6 minutes · Every command and every number below is from the running system.

T-10 minutes — Pre-flight

```
cd server
npm run typecheck && npm test && npm run smoke
```

All three must be clean. Then start it and let it run for **at least 90 seconds** before you present — the picture needs to populate and the posture needs to settle.

```
npm run dev
```

Confirm the baseline:

```
curl -s localhost:3001/api/v1/situation/current | jq '.situation | {threatLevel, threatScore, totalEvents, criticalCount}'
```

Healthy	Unhealthy — do not present
threatLevel: "green" or "yellow"	"red" before you have triggered anything
criticalCount: 0-3	Most events critical
totalEvents: 30-60	0, or hundreds

If unhealthy: `POST /api/v1/simulation/reset` and wait 60 s. If still wrong, a compounding bug has returned — see MASTER_GUIDELINES Part IV.

Have two terminals ready. One running the server, one for curl. Have the map full-screen.

0:00 — Open cold on the live system

"This is a defense operations centre. Five separate data feeds. Radar, perimeter sensors, patrol telemetry, field dispatches — and live weather, which is genuinely coming from Open-Meteo right now, over Ahmedabad."

Do: let the map move. Do not narrate the UI.

"Today, an officer watching this reads five different screens and assembles the picture in their head. VANGUARD assembles it for them — and shows its working."

0:40 — The fusion moment

Do: click a contact with a corroboration badge.

"One contact. Watch what's attached to it."

EV-RAD-T-R-1012	Uncorrelated uav contact	CRITICAL	100%
corroborated by			
EV-LOG-000031	Perimeter trip P-302	430m, 12s	
EV-PER-000037	Visual — GRIZZLY-1	890m, 31s	
EV-INC-000024	Unauthorized entry	1.2km, 48s	

"Radar saw it. A thermal tripwire saw it. A patrol saw it. A dispatcher reported it.
Nobody told the system these were related. It worked that out from distance, time, and which kinds of sensor can confirm which."

Pause here. This is the core claim.

1:20 — Explainability drawer

Do: open the drawer on that contact.

"Every number here decomposes."

```
reliability 0.92 × recency 1.00 × corroboration 1.60 = 100%

sourceReliability 92    dataFreshness    100
spatialAgreement  57    temporalAgreement 93
sourceAgreement   100
```

"Radar is weighted 0.92 because it's calibrated but gets clutter. This reading is one second old, so recency is essentially 1. Four distinct sources agree, which caps the corroboration boost at 1.6."

Then point at the counterfactual:

```
confidence without corroboration: 92%
                                fusion: +8
```

"That's the number that matters. ****Without cross-source corroboration this would be 92%. Fusion bought eight points of certainty.**** And the boost is capped on purpose — otherwise a flood of correlated noise could manufacture confidence out of nothing."

2:10 — Natural language omnibar

Do: type into the omnibar.

"Show me high severity radar contacts near sector 1 in the past hour"

```
Filtering for source radar, severity high/critical, within 60 minutes, near Sector 1 North.
parser: heuristic  latency: 2ms  matches: 2
```

"Two milliseconds. No API call. There's a Gemini parser too, but the deterministic one runs first and always — so the omnibar works with no key and no network."

2:40 — The briefing, and the grounding guarantee

Do: open the briefing panel.

"This is the AI layer. Notice every claim carries event IDs."

Do: click a citation — it opens the raw event.

"Those aren't decorative. Here's why."

Switch to the terminal:

```
curl -s -X POST localhost:3001/api/v1/ai/verify | jq
```

```
{
  "verified": true,
  "totalCitations": 42,
  "uniqueCitations": 24,
  "ungroundedCitations": [],
  "note": "Every cited event ID resolves to a real event in the store."
}
```

"We assume the model is lying. Every ID it produces is resolved against the event store. Invented ones are **stripped**. Claims left with no evidence are **discarded entirely**. That's enforced in code, not requested in a prompt — because a prompt is a request, not a constraint."

Optional, if a judge looks sceptical:

"Pick any citation on that screen and I'll open it."

3:40 — Escalation under load

```
curl -s -X POST localhost:3001/api/v1/simulation/scenario \
-H 'content-type: application/json' \
-d '{"scenario":"border_spike"}' | jq '.expectedEffect'
```

"Coordinated border spike. Seven entry reports across a 3.5 kilometre front, plus two non-squawking fast contacts. Watch the posture."

Do: watch the map. Within ~10 seconds:

```
THREAT    ORANGE (70.9) → RED (147.0) → (221.9)
CRITICAL      2      →      10
```

"The rate detector caught the incident feed running 3.2 sigma above its own baseline. The kinematic detector caught a contact at 312 knots against a 92 knot mean — 3 sigma. The correlation engine formed a multi-source cluster, three distinct feeds agreed, severity escalated, and the posture went red."

Do: open the escalation timeline.

"And it logged exactly which events caused it. 'The system went red' is never an unexplained colour change."

4:30 — Degraded comms

```
curl -s -X POST localhost:3001/api/v1/simulation/degraded \
-H 'content-type: application/json' -d '{"enabled":true}' | jq -r '.effect'
```

"Now I cut the comms."

Do: point at the confidence figures falling across the whole board.

```
mean confidence    93% → 43%
feed reliability 0.92 → 0.37
```

"Most systems would show you an amber dot. Ours drops the **trust weight** of every feed, and that multiplies straight into the confidence formula — so every score across the entire picture falls. The degradation is in the math, not in a banner."

```
curl -s -X POST localhost:3001/api/v1/simulation/degraded \
-H 'content-type: application/json' -d '{"enabled":false}'
```

"Restore, and it comes back to 92."

5:10 — The proof

"Last thing. Every AI project today has an API key. Let me show you what happens when I take ours away."

```
curl -s -X POST 'localhost:3001/api/v1/ai/briefing?deterministic=true' \
| jq '{engine: .summary.provenance.engine, headline: .summary.headline,
      developments: (.summary.keyDevelopments | length),
      coas: (.summary.coursesOfAction | length),
      grounded: .groundingVerified}'
```

```
{
  "engine": "deterministic",
  "headline": "10 critical events active – Uncorrelated uav contact R-1012",
  "developments": 4,
  "coas": 3,
  "grounded": true
}
```

"Full briefing. Four cited developments. Three courses of action with honest tradeoffs.
Zero model calls."

Pause.

"The language model improves the prose. **The fusion engine supplies the substance.** That's the difference between a system and a prompt."

5:50 — Close

"VANGUARD. One picture, every source, zero delay. Purely defensive — no targeting, no weapons release, no lethal autonomy.
Every number on that screen decomposes into factors you can check by hand. Name one and I'll show you where it came from."

Contingencies

If	Then
Weather layer looks empty	It cannot be — cache then synthetic baseline. <i>"That's the fallback doing its job."</i>
Gemini is slow	Briefings are pre-generated and cached; the panel never blocks. Mention it as a design choice.
Posture stuck GREEN	Trigger a scenario. That is what it is for.
Scenario seems to do nothing	Reports drain over several ticks by design. Wait 10 s and narrate the anomaly detectors.
Something genuinely breaks	<code>curl -X POST localhost:3001/api/v1/simulation/reset</code> and keep talking.
Asked for the frontend	Backend is complete and API-first; every panel described is a documented endpoint. Show <code>GET /api/v1</code> .

The four sentences that matter

If you remember nothing else:

1. *"Nobody told the system these were related — it worked it out from space, time, and sensor physics."*
2. *"Without corroboration this would be 92%. Fusion bought eight points."*
3. *"We assume the model is lying, and check every citation in code."*
4. *"Take the API key away and it still works. The fusion engine is the product."*